Proofs and Problem Solving

Week 2: Inequalities and Induction

Notes based on Martin Liebeck's

A Concise Introduction to Pure Mathematics

Contents

II Week 2: Inequalities and Induction

3. Inequalities

This week we will examine inequalities more closely, and describe them formally in terms of just a few rules or *axioms*.

3.1 The axioms for "<"

An inequality is a relation[1] on pairs of real numbers. We use the symbols $\leqslant, \geqslant, <, >$ to denote these relationships. E.g. we write $x < y$, spoken as "x is less than y". The informal idea is that $x < y$ exactly when "x is strictly to the left of y on the number line". The inequality $x < y$ means exactly the same thing as $y > x$ (spoken as "y is greater than x"). When we write $x \leqslant y$ (spoken as "x is less than or equal to y" or "x is at most y") we mean that either $x < y$ or $x = y$. Hence it is true that $2 \leqslant 3$. We prefer to list the formal properties that $<$ possesses which, together with the rules of addition and rules of multiplication from Section 2, make $\mathbb{R}$ into what is called an *ordered field*. These properties of the relation $<$ on the real numbers should be very familiar!

Definition 3.1 An *ordered field* is a field F, together with a relation $<$, so that the following hold:

(R1) (Positive/negative) If $x \in F$, then exactly one of the following is true: $0 < x$, $x = 0$, or $x < 0$.
(R2) (Negations reverse) If $y < x$, then $-x < -y$.
(R3) (Can add constants) If $x < y$, and $c \in F$, then $x + c < y + c$.
(R4) (Positives multiply) If $0 < x$ and $0 < y$ then $0 < xy$.
(R5) (Transitive property) If $x < y$ and $y < z$ then $x < z$.

This definition gives us a set of *axioms*. That is, rules that we will take for granted as a starting point. The mathematical community are careful to choose a *minimal* set of independent axioms in the sense that no axiom can be deduced from the others. But is the

[1] We formally define a relation in Week 8 of PPS. For now, just think of a relation as a way of classifying pairs of objects *e.g* we could say two people are related if they go to the same university.

set of axioms in Definition 3.1 minimal? In this section we'll show how many standard facts about inequalities that we are used to can be deduced from these axioms.

Proposition 3.1 If $x > 0$, then $-x < 0$.

Proof. This follows immediately from (R2) with 0 in place of y. ■

Proposition 3.2 If $x \neq 0$, then $x^2 > 0$.

Proof. If $x > 0$ this follows from (R4). If $x < 0$ then $-x > 0$, by Proposition 3.1 so, using (R4),

$$0 < (-x)^2 = (-1)^2 x^2 = x^2.$$

(Recall from Exercise 2.1 that $(-1)^2 = 1$.) ■

■ **Exercise 3.1** Prove that $1 > 0$.

Pause for thought: why are we "wasting our time" proving things like $1 > 0$, which we have all known and understood since the early years of primary school? Mathematics is not just to establish what is true and what is false, but also to understand, in a critical way, the relationships between statements. What we learn from this exercise is that the statement $1 > 0$ can be logically derived from the basic axioms (A0-A4), (M0-M5) and (R1-R5). The same applies to most other true statements of elementary arithmetic.

Proposition 3.3 If $x > 0$, then $\frac{1}{x} > 0$.

Proof. Suppose $\frac{1}{x} \leqslant 0$. Then $-\frac{1}{x} \geqslant 0$, and so (R4) implies

$$0 \leqslant -\frac{1}{x} \cdot x = -1 < 0,$$

which is a contradiction, thus $\frac{1}{x} > 0$. ■

Proposition 3.4 If $x > 0$, then $u > v$ if and only if $xu > xv$.

Proof. First assume $u > v$. If $x > 0$, then

$$\begin{aligned} u > v &\overset{[R3]}{\Longrightarrow} u - v > v - v = 0 \\ &\overset{[R4]}{\Longrightarrow} (u-v)x > 0 \\ &\Longrightarrow ux - vx > 0 \\ &\overset{[R3]}{\Longrightarrow} ux > vx \end{aligned}$$

Now assume $xu > xv$. Then we can use the same proof as above, but replacing u with ux, v with vx, and x with $\frac{1}{x}$ (since $\frac{1}{x} > 0$ by Proposition 3.3). ■

Proposition 3.5 If $0 < u, v$, then $u^2 > v^2$ if and only if $u > v$.

Proof. First, notice that if $0 < v, u$, then

$$u^2 > v^2 \quad \overset{[R3]}{\Longleftrightarrow} \quad u^2 - v^2 > v^2 - v^2 = 0 \quad \Longleftrightarrow \quad (u-v)(u+v) > 0$$

and by (R4), this last inequality is true if and only if $u - v > 0$ (since $u + v > 0$ by assumption), which by (R3) is true if and only if $0 + v < u - v + v$, that is, if and only if $v < u$. ■

3.2 Separating products

In this section we cover some very useful inequalities that allow us to essentially separate terms in a product. They all stem from the following lemma:

Lemma 3.6 For $u, v \in \mathbb{R}$,

$$uv \leqslant \frac{u^2+v^2}{2}.$$

Notice that on one side of the above inequality, the terms u and v are multiplied together, and on the other side, they are separate (but now squared).

Proof. Since$(u-v)^2 \geqslant 0$, we get

$$0 \leqslant (u-v)^2 = u^2 - 2uv + v^2.$$

So by (R3), we can add $2uv$ to both sides to get

$$2uv \leqslant u^2 + v^2$$

and then by (R4), we can multiply both sides by $\frac{1}{2}$ to get $uv \leqslant \frac{u^2+v^2}{2}$ as desired. ∎

▪ **Exercise 3.2** When does equality hold in the previous lemma? That is, when do we have $uv = \frac{u^2+v^2}{2}$?

From this, we get the famous Arithmetic Mean-Geometric Mean (AM-GM) inequality:

Theorem 3.7 — AM-GM Inequality. Let $n \in \mathbb{N}$ and $x_1, \ldots, x_n \geqslant 0$. Then

$$(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} \leqslant \frac{x_1 + \cdots + x_n}{n}.$$

That is, the *geometric mean* on the left is at most the *arithmetic mean* on the right.

Proof. The $n = 2$ case follows from Lemma 3.6: if we set $u = \sqrt{x_1}$ and $v = \sqrt{x_2}$, then

$$(x_1 x_2)^{\frac{1}{2}} = uv \leqslant \frac{u^2+v^2}{2} = \frac{x_1+x_2}{2}. \tag{3.2.1}$$

For the $n > 2$ case, see Exercise 4.11. ∎

▪ **Exercise 3.3** For $x_1, x_2 \geqslant 0$, when do we have $(x_1 x_2)^{\frac{1}{2}} = \frac{x_1+x_2}{2}$?

The next useful inequality is the *Cauchy-Schwarz* inequality:

Theorem 3.8 — Cauchy-Schwarz Inequality. Let $n \in \mathbb{N}$, $x_1, \ldots, x_n, y_1, \ldots, y_n \in \mathbb{R}$. Then

$$x_1 y_1 + \cdots + x_n y_n \leqslant \sqrt{x_1^2 + \cdots + x_n^2}\sqrt{y_1^2 + \cdots + y_n^2}.$$

Why is this useful? On the left we have a *mixed* product of x-terms and y-terms, and we are able to relate it to a product of two terms where now the x's and y's are *separated*.

We'll just prove the $n = 2$ case for now and return to the general case later. Note that

$$(x_1 y_1 + x_2 y_2)^2 = x_1^2 y_1^2 + x_2^2 y_2^2 + 2x_1 y_1 x_2 y_2$$

and using Lemma 3.6 on the second term with $u = x_1 y_2$ and $v = x_2 y_1$, this above is at most

$$\leqslant x_1^2 y_1^2 + x_2^2 y_2^2 + 2\frac{(x_1 y_2)^2 + (x_2 y_1)^2}{2} = x_1^2 y_1^2 + x_2^2 y_2^2 + x_1^2 y_2^2 + x_2^2 y_1^2 = (x_1^2 + x_2^2)(y_1^2 + y_2^2)$$

Thus we have shown

$$(x_1y_1+x_2y_2)^2 \leqslant (x_1^2+x_2^2)(y_1^2+y_2^2) = \left(\sqrt{x_1^2+x_2^2}\sqrt{y_1^2+y_2^2}\right)^2$$

and now by Proposition 3.5, we get

$$x_1y_1+x_2y_2 \leqslant \sqrt{x_1^2+x_2^2}\sqrt{y_1^2+y_2^2}.$$

Example 3.1 Show that

$$\frac{a+b}{2} \leqslant \left(\frac{a^2+b^2}{2}\right)^{\frac{1}{2}}.$$

Which of our above inequalities should we try using? Lemma 3.6 and AM-GM don't seem relevant since we don't have something like ab on the left, so let's see if we can use the Cauchy-Schwarz inequality: Rewrite $\frac{a+b}{2} = a\cdot\frac{1}{2}+b\cdot\frac{1}{2}$, then the Cauchy Schwarz inequality implies:

$$\frac{a+b}{2} = a\cdot\frac{1}{2}+b\cdot\frac{1}{2} \leqslant \sqrt{a^2+b^2}\sqrt{\frac{1}{2^2}+\frac{1}{2^2}} = \left(\frac{a^2+b^2}{2}\right)^{\frac{1}{2}}.$$

3.3 Absolute Values

Definition 3.2 For $x \in \mathbb{R}$, we define the *absolute value* or *modulus* of x to be

$$|x| = \begin{cases} x & \text{if } x \geqslant 0 \\ -x & \text{if } x \leqslant 0 \end{cases}$$

Geometrically, $|x|$ measures the distance from x to 0 along the real line, e.g. $|-3| = 3$.

Lemma 3.9 For $x \in \mathbb{R}$,

$$-|x| \leqslant x \leqslant |x|.$$

Proof. If $x \geqslant 0$, then since $|x| \geqslant 0$,

$$-|x| \leqslant 0 \leqslant x = |x|$$

and if $x \leqslant 0$, then $-x = |x|$, so $x = -|x|$, hence

$$-|x| = x \leqslant 0 \leqslant |x|.$$

■

Tip **Proofs with $|\cdot|$:** Since the definition of $|x|$ is defined in terms of cases, a proof involving absolute values may need to be split into cases depending on what sign x has.

Example 3.2 For $x \in \mathbb{R}$, $|x|^2 = x^2$.

Proof. We split the proof into two cases.

Case 1: If $x \geqslant 0$. Then $|x| = x$ and so $|x|^2 = x^2$, which proves the claim in this case.
Case 2: If $x < 0$, then $|x| = -x$, so $|x|^2 = (-x)^2 = (-1)^2x^2 = 1$.
Thus, the claim is true in all cases.

■

Example 3.3 For which $x \in \mathbb{R}$ do we have $|x-1| < |x+2|$?

Notice that the signs of the absolute value change as x crosses 1 and -2, so we split into cases depending on where x lies in relation to these numbers:

Case 1: If $x \leqslant -2$, then $x-1 < 0$, so $|x-1| = 1-x$ and $x+2 \leqslant 0$ so $|x+2| = -2-x$, so $|x-1| < |x+2|$ holds if and only if $1-x < -2-x$, or equivalently, $2 < -1$, which never holds.

Case 2: If $-2 \leqslant x \leqslant 1$, then $|x-1| = 1-x$ again but $|x+2| = x+2$, so $|x-1| < |x+2|$ if and only if $1-x < x+2$, i.e. if $-1 < 2x$, so the only x's between -2 and 1 where the inequality holds is for $x > -\frac{1}{2}$.

Case 3: If $x \geqslant 1$, then $|x-1| = x-1$ and $|x+2| = x+2$, so $|x-1| < |x+2|$ if and only if $x-1 < x+2$, i.e. if $-1 < 2$, which always holds.

Bringing it all together, we see that our inequality holds if and only if $-\frac{1}{2} < x$.

Lemma 3.10 For $x,y \in \mathbb{R}$, $|xy| = |x| \cdot |y|$.

We leave the proof as an exercise.

Theorem 3.11 — The Triangle Inequality. For $x,y \in \mathbb{R}$,

$$|x+y| \leqslant |x|+|y|.$$

Proof. We split into cases, and will use the fact that $\pm x \leqslant |x|$ by definition.

Case 1: If $x+y \geqslant 0$, then

$$|x+y| = x+y \leqslant |x|+|y|.$$

Case 2: If $x+y \leqslant 0$, then

$$|x+y| = -(x+y) = -x+(-y) \leqslant |x|+|y|.$$

Thus, the claim is true in all cases.

■

We leave the following corollary as an exercise.

Corollary 3.12 — The Reverse Triangle Inequality. For $x,y \in \mathbb{R}$,

$$\big||x|-|y|\big| \leqslant |x+y|.$$

In particular,

$$|x|-|y| \leqslant |x+y|.$$

3.4 Exercises

■ **Exercise 3.4** For all $a,b \in \mathbb{R}$, show that

$$-\max\{a^2,b^2\} \leqslant ab \leqslant \max\{a^2,b^2\}.$$

■ **Exercise 3.5** Show that for all $a,b,c \in \mathbb{R}$ we have $|a-c| \leqslant |a-b|+|b-c|$.

■ **Exercise 3.6** For all $a,b \in \mathbb{R}$, show that $|a+b| \geqslant |a|-|b|$. In fact, show that $|a+b| \geqslant ||a|-|b||$.

■ **Exercise 3.7** For $a,b>0$, show that

$$\sqrt{ab} \geqslant \frac{2ab}{a+b}.$$

■ **Exercise 3.8** Show that if $0<a\leqslant b$ and $a,b\in\mathbb{R}$, then

$$a \leqslant \frac{a+b}{2} \leqslant b.$$

Show that either of these inequalities is an equality if and only if $a=b$.

■ **Exercise 3.9** Show that if $a,b,c,d>0$ and $0<\frac{a}{b}<\frac{c}{d}$, then

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

■ **Exercise 3.10** Show the following for $a,b,c,d \in \mathbb{R}$:
(a) $(a^2-b^2)(c^2-d^2) \leqslant (ac-bd)^2$
(b) $(a^2+b^2)(c^2+d^2) \geqslant (ac+bd)^2$.
(c) $(a^2-b^2)^2 \geqslant 4ab(a-b)^2$.

■ **Exercise 3.11** Show that for $a,b\in\mathbb{R}$ and $d>0$,

$$ab \leqslant \frac{a^2}{d}+2db^2.$$

Hint: First $ab = \frac{a}{c}(bc)$.

■ **Exercise 3.12** Show that for $a>0$, $a+\frac{1}{a}\geqslant 2$.

■ **Exercise 3.13** Suppose $x,y>0$ and r is a rational number so that $0<r\leqslant 1$. Use the AM-GM inequality to prove

$$x^r y^{1-r} \leqslant rx+(1-r)y.$$

■ **Exercise 3.14** **(Challenging!)** Prove that if $m_1,\ldots,m_k$ are positive integers, $n=m_1+m_2+\cdots+m_k$ and $y_1,\ldots,y_k>0$, then

$$\frac{m_1y_1+m_2y_2+\cdots+m_ky_k}{m_1+m_2+\cdots+m_k} \geqslant \sqrt[n]{y_1^{m_1}\cdot y_2^{m_2}\cdots y_k^{m_k}}$$

■ **Exercise 3.15** Given that $x,y,z\in\mathbb{N}$, solve

$$\left(1+\frac{1}{x}\right)\left(1+\frac{1}{y}\right)\left(1+\frac{1}{z}\right)=3.$$

4. Induction

Mathematical induction is a particular form of proof. The goal of a proof by induction is to show that each member of an infinite family of statements is true.

4.1 Preliminaries on Σ notation

Before getting into induction, it will be useful to review Σ notation. Σ notation is spoken as "sigma notation" after the Greek letter Σ.

Definition 4.1 Given a sequence of numbers $a_m, a_{m+1}, \ldots, a_n$, we write

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + \cdots + a_n.$$

For example,

$$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + 3^2 + \cdots + n^2.$$

We can split sums and use the distribution

$$\sum_{k=m}^{n} (a_k + b_k) = \sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k \quad \text{and} \quad \sum_{k=m}^{n} c \cdot a_k = c \cdot \sum_{k=m}^{n} a_k.$$

Note: this is why we sometimes call numbers *constants*: the number c when it appears inside a sum does not vary as k changes, and for this reason we can factor it outside the sum.

The following are useful manipulations of Σ notation

1. Detaching the last element:

$$\sum_{k=1}^{n+1} a_k = \left(\sum_{k=1}^{n} a_k\right) + a_{n+1}.$$

2. Differences
$$\text{If } S_n = \sum_{k=1}^{n} a_k, \text{ then } S_{n+1} - S_n = a_{n+1}.$$
3. Change of local variable:
$$\sum_{k=m}^{n} a_k = \sum_{j=m}^{n} a_j.$$
4. Shift of local variables:
$$\sum_{k=m}^{n} a_{k+\ell} = \sum_{k=m+\ell}^{n+\ell} a_k.$$
For example,
$$\sum_{k=3}^{5} \frac{1}{(k+1)^2} = \frac{1}{(3+1)^2} + \frac{1}{(4+1)^2} + \frac{1}{(5+1)^2} = \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} = \sum_{k=4}^{6} \frac{1}{k^2}.$$

4.2 Induction

The main topic of this chapter is the following fundamental property of the natural numbers/axiom of mathematics called the Principle of Mathematical Induction.

Definition 4.2 — The Principle of Mathematical Induction. Given a list of statements $P(k), P(k+1), \ldots$, we may conclude that $P(n)$ is true for every integer $n \geqslant k$ *provided that*
- we know that $P(k)$ is true ("base case")
- and we can prove that $P(n) \Rightarrow P(n+1)$ for any integer $n \geqslant k$ ("induction step").

The basic idea of an induction proof is to proceed step by step. We prove the base case is true, and the induction step retains truth as we move from $P(n)$ to $P(n+1)$.

A full written proof by induction has the following four parts clearly separated.
1. A clear and explicit statement of the “induction hypothesis” $P(n)$ and the intention to prove that $P(n)$ is true for all $n \geqslant k$ by induction.
2. Prove a “base case” is true. Typically we prove the single statement $P(k)$.
3. Prove the “induction step”. Typically, we prove if $P(n)$ is true then $P(n+1)$ is also true.
4. Conclude that steps (2) and (3) allow the conclusion that $P(n)$ is true for all natural numbers $n \geqslant k$ by the principle of mathematical induction.

Mathematics written for experts might skimp on some of the detail, e.g. saying “ the base case is trivial”. In this course your proofs must be written systematically using all four parts of the format described above.

Example 4.1 For $n \in \mathbb{N}$, $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$.

Proof. Let $P(n)$ be the statement that $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. We prove that $P(n)$ is true for all $n \in \mathbb{N}$ by induction as follows:

- **Base Case:** First let's prove $P(1)$. This is just
$$\sum_{k=1}^{1} k = 1 = \frac{1(1+1)}{2}.$$

Thus, the base case is true.

- **Induction Step:** Assume $P(n)$ holds and consider

$$\sum_{k=1}^{n+1} k = \left(\sum_{k=1}^{n} k\right) + (n+1) = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2},$$

where we used the statement $P(n)$ to replace $\sum_{k=1}^{n} k$ with $\frac{n(n+1)}{2}$. Hence $\sum_{k=1}^{n+1} k = \frac{(n+1)(n+2)}{2}$ which proves that $P(n) \Rightarrow P(n+1)$.

Since we have shown the base case and induction step, $P(n)$ is true for all $n \in \mathbb{N}$ by induction. ■

■ **Exercise 4.1** Prove the following formulas hold for every $n \in \mathbb{N}$ by induction:

(a) $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

(b) $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$.

It turns out that for all powers m there is a degree $m+1$ polynomial that gives a formula for the value of $\sum_{k=1}^{n} k^m$. To learn more, investigate *Bernoulli numbers*.

Some statements may not be true for all $n \geqslant 1$ but maybe for $n \geqslant k$ for some k, and depending on the problem you will need to figure out what this k is.

Example 4.2 For which $n \in \mathbb{N}$ do we have $2^n < n!$? If we plug in $n = 1,2,3$ this inequality fails, but for $n \geqslant 4$ it starts to hold. This gives rise to the following conjecture:

Claim: $2^n < n!$ for $n \geqslant 4$.

Proof. We prove by induction. Let $P(n)$ be the statement that $2^n < n!$.

- **Base case:** The base case is $n = 4$, and it holds because $2^4 = 16 < 4! = 24$.

- **Induction Step:** Assume $P(n)$ is true, i.e. $2^n < n!$ for some $n \geqslant 4$, then

$$2^{n+1} = 2^n \cdot 2 < n! \cdot 2 < n! \cdot (n+1) = (n+1)!.$$

Since we have shown the base case and induction step, $P(n)$ is true for all $n \geqslant 4$ by induction. ■

Some problems can be proven with or without induction:

■ **Exercise 4.2** Prove that $n^3 - n$ is divisible by 6 for all $n \in \mathbb{N}$, first by induction, then without induction.

The following formula will be very useful throughout this course:

Theorem 4.1 — Geometric Series Formula. Let $x \neq 1$ be a real number. Then for $N \in \mathbb{N}$,

$$\sum_{n=1}^{N} x^n = \frac{x - x^{N+1}}{1-x} \quad \text{and} \quad \sum_{n=0}^{N} x^n = \frac{1 - x^{N+1}}{1-x} \tag{4.2.1}$$

Proof. The second equation follows from the first (just by adding 1), so we just need to prove the first. We prove this by induction.

- **Base case:** If $N = 1$, then

$$\frac{x - x^2}{1 - x} = \frac{x(1-x)}{1-x} = x = \sum_{n=1}^{1} x^n.$$

- **Induction Step:** Suppose (4.2.1) holds for some N. Then

$$\sum_{n=1}^{N+1} x^n = x^{N+1} + \sum_{n=1}^{N} x^n = x^{N+1} + \frac{x - x^{N+1}}{1-x} = \frac{x^{N+1} - x^{N+2}}{1-x} + \frac{x - x^{N+1}}{1-x} = \frac{x - x^{N+2}}{1-x}.$$

This proves the induction step. Thus, (4.2.1) holds for all $N \in \mathbb{N}$ by induction.

■

4.3 Strong Mathematical Induction

Sometimes it's not enough to know that $P(n)$ is true in order to prove that $P(n+1)$ is true; we might additionally need to use that some of the "previous" statements $P(k), P(k+1), \ldots, P(n-1)$ are true.

Definition 4.3 — Principle of Strong Mathematical Induction. Let $k \in \mathbb{N}$ and let $P(k), P(k+1), \ldots$ be statements. Suppose that

- $P(k)$ is true, ("base case" - may require more than one)
- for any integer $n \geqslant k$, statements

$$P(k), P(k+1), P(k+2), \ldots, P(n) \text{(all taken together)} \implies P(n+1)$$

("induction step").

Then $P(n)$ is true for every positive integer $n \geqslant k$.

Strong induction is needed when you need to make use of several previous statements to prove the next one. For example:

Theorem 4.2 Every integer $n \geqslant 2$ can be written as a product of primes $n = p_1 \cdots p_k$.

If n is prime, this statement still makes sense: we just interpret n as being the product of just one number, n itself.

Proof. We prove by strong induction on n. The case $n = 2$ immediately holds (taking into account the previous remark). For the induction step, suppose the theorem holds for all integers $n < N$ (this is our strong inductive hypothesis). If N is prime, there is nothing to prove; otherwise, if N is not prime, then $N = a \cdot b$ for some positive integers a and b greater than 1 and smaller than N. Both a and b can be decomposed by the strong induction hypothesis, thus so can $N = ab$. ■

Another situation when you need strong induction is when studying recurrence relations. A *recurrence* relation is a sequence $a_1, a_2, \ldots$ of numbers where each a_n is defined in terms of previous terms in the sequence. The most famous recurrence relation is the *Fibonacci sequence*, which is defined as $f_1 = 1$, $f_2 = 1$ and for $n > 2$, $f_n = f_{n-1} + f_{n-2}$.

Example 4.3 Let $x_1 = 1, x_2 = 3$ and for $n \geqslant 2$, let

$$x_{n+1} = 4x_n - 3x_{n-1}.$$

Let's try and find a general pattern for what x_n is. If we test the first few values, we get

$$x_1 = 1,\ x_2 = 3,\ x_3 = 4\cdot 3 - 3\cdot 1 = 9,\ x_4 = 4\cdot 9 - 3\cdot 3 = 27$$

so it seems like $x_n = 3^{n-1}$. We will prove this by strong induction. Let $P(n)$ be the statement "$x_n = 3^{n-1}$". Let's first prove the first two base cases: $x_1 = 1 = 3^{1-1}$ and $x_2 = 3 = 3^{2-1}$, so $P(1)$ and $P(2)$ are true. Now suppose $P(k)$ is true for $k = 1, 2, \dots, n$ for some $n \geqslant 2$. Then

$$x_{n+1} = 4x_n - 3x_{n-1} = 4\cdot 3^{n-1} - 3x_{n-1} = 4\cdot 3^{n-1} - 3\cdot 3^{n-2} = 3^n = 3^{(n+1)-1}.$$

This proves the induction step, and thus $x_n = 3^{n-1}$ for all n by strong induction.

Discussion: *Why did we prove two base cases?* Note that the formula $x_{n+1} = 4x_n - 3x_{n-1}$ holds only for $n \geqslant 2$, so in particular, I can't use it to prove $P(2)$ (i.e. that $x_2 = 3^{2-1}$). For this reason, I need to prove a few more base cases that can't be covered in my induction step.

Why did we need strong induction instead of induction? I needed both $P(n)$ and $P(n-1)$ to be true to prove $P(n+1)$, but with standard induction I would need to prove $P(n+1)$ follows from just $P(n)$.

Tip *How do I know when to use induction vs. strong induction?* One way is to first try proving the induction step to see how many previous cases you need, then that will tell you how many base cases you need as well: if it is more than one, then you need strong induction.

4.4 Inequalities revisited

Powered with induction, we can now prove some more useful inequalities.

Proposition 4.3 Suppose $x, y > 0$ and $n \in \mathbb{N}$.
(a) $x^n > y^n$ if and only if $x > y$.
(b) $x^{\frac{1}{n}} > y^{\frac{1}{n}}$ if and only if $x > y$.

Proof. Let $x, y > 0$.
(a) First we show $x > y$ implies $x^n > y^n$. We prove this by induction. The $n = 1$ case is trivial. Suppose the claim is true for some integer n, we will prove the $n+1$ case. Let $x > y > 0$. Then

$$x^{n+1} - y^{n+1} = y^{n+1}\left(\left(\frac{x}{y}\right)^{n+1} - 1\right).$$

Let $a = \frac{x}{y}$. Since $x > y > 0$, Proposition 3.3 implies $1/y > 0$ and Proposition 3.4 implies $x \cdot \frac{1}{y} > y \cdot \frac{1}{y}$, that is, $\frac{x}{y} > 1$. Letting $a = \frac{x}{y}$,

$$x^{n+1} - y^{n+1} = y^{n+1}\left(a^{n+1} - 1\right) = y^{n+1}(a^n a - 1) \geqslant y^{n+1}(a^n \cdot 1 - 1) = y^{n+1}(a^n - 1).$$

By the induction hypothesis, we know $x^n > y^n$, which implies $a^n = \frac{x^n}{y^n} > 1$, so the above quantity is positive. Thus, $x > y > 0$ implies $x^n > y^n$.
The converse statement is $x^n > y^n$ implies $x > y$, which is equivalent to the contrapositive statement that $x \leqslant y$ implies $x^n \leqslant y^n$. If $x = y$, then clearly $x^n = y^n$; if $x < y$, then $x^n < y^n$ follows from the above proof.

(b) By part (a), $x^{\frac{1}{n}} > y^{\frac{1}{n}}$ if and only if $(x^{\frac{1}{n}})^n > (y^{\frac{1}{n}})^n$, that is, $x > y$.

■

Let's finish proving the Cauchy-Schwarz inequality:

$$x_1y_1 + \cdots + x_ny_n \leqslant \sqrt{x_1^2 + \cdots + x_n^2}\sqrt{y_1^2 + \cdots + y_n^2} \quad \text{for} \quad x_1, \ldots, x_n, y_1, \ldots, y_n \in \mathbb{R}.$$

We prove this by Strong Induction. We already proved the base case $n = 2$. Suppose we have shown the above inequality for $n = 2, 3, \ldots, N$. By applying the $n = N$ and $n = 2$ cases in succession,

$$\begin{aligned}
x_1y_1 + \cdots + x_Ny_N + x_{N+1}y_{N+1} &\leqslant \sqrt{x_1^2 + \cdots + x_N^2}\sqrt{y_1^2 + \cdots + y_N^2} + x_{N+1}y_{N+1} \\
&\leqslant \left(\sqrt{x_1^2 + \cdots + x_N^2}^2 + x_{N+1}^2\right)^{\frac{1}{2}} \left(\sqrt{y_1^2 + \cdots + y_N^2}^2 + y_{N+1}^2\right)^{\frac{1}{2}} \\
&= \left(x_1^2 + \cdots + x_N^2 + x_{N+1}^2\right)^{\frac{1}{2}} \left(y_1^2 + \cdots + y_N^2 + y_{N+1}^2\right)^{\frac{1}{2}}
\end{aligned}$$

This proves the induction step, and thus proves the Cauchy-Schwarz inequality.

4.5 Exercises

■ **Exercise 4.3** (Aug 2018 Exam) Show that, for all $n \geqslant 0$, $\sum_{k=0}^{n} k \cdot k! = (n+1)! - 1$.

■ **Exercise 4.4** Show that for all $n \geqslant 0$, $\sum_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$.

■ **Exercise 4.5** Show that $n! \leqslant n^n$ for all $n \geqslant 1$

■ **Exercise 4.6** Suppose $x_1 = 1$ and $x_{n+1} = \sqrt{1 + 2x_n}$ for all $n \geqslant 1$. Show $x_n \leqslant 4$ for all n.

■ **Exercise 4.7** Let $f_1 = f_2 = 1$ and $f_n = f_{n-1} + f_{n-2}$ for $n > 2$.
(a) Show that f_{4n} is divisible by 3 for all $n \geqslant 1$.
(b) Show that $1 \leqslant f_{n+1}/f_n \leqslant 2$ for all $n \geqslant 1$.

■ **Exercise 4.8** Find a formula for the number of diagonals in a convex polygon with $n \geqslant 3$ vertices.

■ **Exercise 4.9** Show that for all integers $n \geqslant 3$, there are *distinct* integers $x_1, \ldots, x_n$ so that

$$\sum_{k=1}^{n} \frac{1}{x_k} = 1.$$

(It is an open question whether you can always do this with x_n distinct *odd* numbers.)

■ **Exercise 4.10** Joseph Bertrand conjectured in 1845 (and it was proven by Chebychev in 1852) that, if $p_1, p_2, \ldots$ are the primes in ascending order, then

$$p_{n+1} < 2p_n.$$

This is called **Bertrand's postulate**. Using this result, show that for all $n \geqslant 4$ we have

$$p_n < \sum_{k=1}^{n-1} p_k.$$

■ **Exercise 4.11** **Challenge:** In this exercise, we will prove the Arithmetic-Geometric Inequality: for $x_1, \dots, x_n \geqslant 0$,

$$(x_1 \cdots x_n)^{\frac{1}{n}} \leqslant \frac{x_1 + \cdots + x_n}{n}.$$

We will do this in several steps.

(a) Prove that for all $n \in \mathbb{N}$, if $x_1, \dots, x_{2^n}$ are positive numbers, then

$$(x_1 x_2 \cdots x_{2^n})^{1/2^n} \leqslant \frac{x_1 + \cdots + x_{2^n}}{2^n}.$$

(b) Use the previous case to prove the AM-GM inequality. *Hint: Let* $2^m > n$ *and define a new sequence* $y_1, \dots, y_{2^m}$ *with* $y_i = x_i$ *for* $i = 1, 2, \dots, n$ *and* $y_i = A$ *otherwise where* $A = \frac{x_1 + \cdots + x_n}{n}$. *Apply the previous case to the product of the* y_i *and find an appropriate choice of* A *that will imply the inequality for the* x_i.